

ICT 1209 – Web Technologies

Mini Project: Final Report

NiRu Furniture

Furniture showcase and order management system

Group Memebers

Name	Registration Number	Index
Nisala Anjana	ITT/2024/027	2712
Sandeesh Rukshan	ITT/2024/092	2777

Department of ICT
Faculty of Technology
Rajarata University of Sri Lanka

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1. PROJECT OVERVIE

1.1 Introduction

NiRu Furniture is a database driven web application developed as an interactive furniture showcase and order management system. The system is designed to provide customers with a convenient platform where they can browse furniture products, search and filter products, view detailed product information, manage their accounts, maintain Wishlist, and submit and monitor orders.

The application also provides an administrative interface through which administrators can manage products, customers, orders, inventory information, and other important aspects of the system.

The project combines frontend web technologies with server-side programming and database management. The system uses HTML5, CSS3, Bootstrap 5, JavaScript, PHP, and MySQL in accordance with the technology requirements of the ICT 1209 Web Technologies mini project. The official project guide specifies HTML5/CSS3/Bootstrap 5 for frontend structure, Vanilla JavaScript for client-side logic, PHP 8 for backend, MySQL for the database, and Git/GitHub for version control.

1.2 Purpose of the project

The main purpose of NiRu furniture is to develop a modern and user friendly furniture management web application that improves the way customers interact with the furniture business.

The system allows customers to:

- Browse available furniture collections.
- Search for furniture products.
- Filter products according to categories.
- View detailed product information.
- Register for an account.
- Log into the system securely.
- Add products to their wishlist.
- Manage their orders.
- View order history.
- Update personal information.
- Contact the furniture business.

The system also provides administrators with facilities to:

- Manage furniture products.
- Add new products.
- Update existing products.
- Delete products.
- Manage inventory.
- Update order statuses.
- View customer information.
- Monitor sales-related information.

These functions provide both customers and administrators with a centralized platform for interacting with and managing the furniture business.

1.3 Target Audience

The main target users of the system are:

Homeowners

Homeowners can use the system to browse furniture collections and identify products suitable for their homes.

Office Owners

Office owners can browse products and identify furniture suitable for office environments.

Interior Designers

Interior designers can use the product catalogue to explore furniture options for different interior projects.

Furniture Retailers

Furniture retailers can use the platform to explore and manage furniture products and orders.

Furniture Businesses

Furniture businesses can use administrative functions to manage their products, customers, and orders.

1.4 Project Objective

The main objectives of the project are:

1. To develop a database-driven furniture web application.
2. To provide an attractive and user-friendly interface.
3. To allow customers to browse and search furniture products.
4. To implement user registration and authentication.
5. To provide customers with personalized dashboards.
6. To provide Wishlist and order-management functionality.
7. To create an administrator dashboard.
8. To allow administrators to perform product management operations.
9. Managing customer orders through the system.
10. To connect the frontend with a MySQL database using PHP.
11. To develop a responsive interface suitable for different screen sizes.
12. To apply appropriate input validation and data-sanitization techniques.

1.5 Key Features

The major features of the system include:

Customer Features

- Home page
- Product catalogue
- Product search
- Category filtering
- Product details
- User registration
- User login/logout
- Customer dashboard
- Order history
- Wishlist
- Account settings
- Password management
- Contact form

Administrator Features

- Admin dashboard
- Product management
- Add product
- Edit product
- Delete product
- Inventory management
- Order management
- Order status management
- Customer management
- User-role management
- Sales-related information

The project overview and feature set are consistent with the submitted project documentation, which describes the customer and administrator functions in detail.

2.SYSTEM DESIGN

2.1 System Overview

NiRu Furnitures follows a client-server web application structure.

The front end provides the user interface through which customers and administrators interact with the application. PHP is used to process requests and communicate with the MySQL database. MySQL stores important information such as users, products, orders, and Wishlist data.

The general flow of the system can be represented as:

User → Web Interface → PHP Backend → MySQL Database → PHP Backend → Web Interface

When a user performs an operation such as logging in or viewing a product, the request is processed by the PHP backend. The backend communicates with the database when necessary and returns the appropriate information to the user interface.

1.3 User Roles

The system contains two main user roles.

Customer

Customers can:

- Register an account.

- Log in.
- Browse products.
- Search products.
- Filter products.
- View product details.
- Add products to their Wishlist.
- Manage their profile.
- View orders.
- Track order status.

Administrator

Administrators have additional management privileges.

Administrators can:

- Manage products.
- Manage inventory.
- View customer information.
- View orders.
- Update order statuses.
- Manage user roles.
- Monitor system information.

Role-based access is important because normal customers should not be able to access administrative functionality.

2.3 Customer-Side Structure

The customer section consists of several interconnected pages.

Home

The home page introduces the furniture business and provides navigation to the main areas of the website.

Shop

The shop page provides the main product catalogue.

Product Details

The product details page provides detailed information about a selected furniture item.

Authentication

Customers can create accounts and log into the system.

Customer Dashboard

The dashboard provides customers with access to their orders, wishlist, settings, and profile information.

Contact

The contact page allows customers to send messages to the business.

The submitted report describes the customer-side functionality including Home, Shop, Product Details, Authentication, Dashboard, Orders, Wishlist, Settings, and Contact pages.

2.4 Administrator-Side Structure

The administrator section provides centralized system management.

The main administrative functions are:

- Dashboard
- Products
- Orders
- Users
- Settings

The admin dashboard contains KPI information such as total products, total orders, total customers, and total revenue. It also provides a recent-orders management table.

2.5 Database Design

The application uses a MySQL relational database.

The database is responsible for storing application information and maintaining relationships between different types of data.

The major database entities include:

- Users
- Products
- Categories
- Orders
- Wishlist

The database design allows the application to maintain information efficiently while reducing unnecessary duplication.

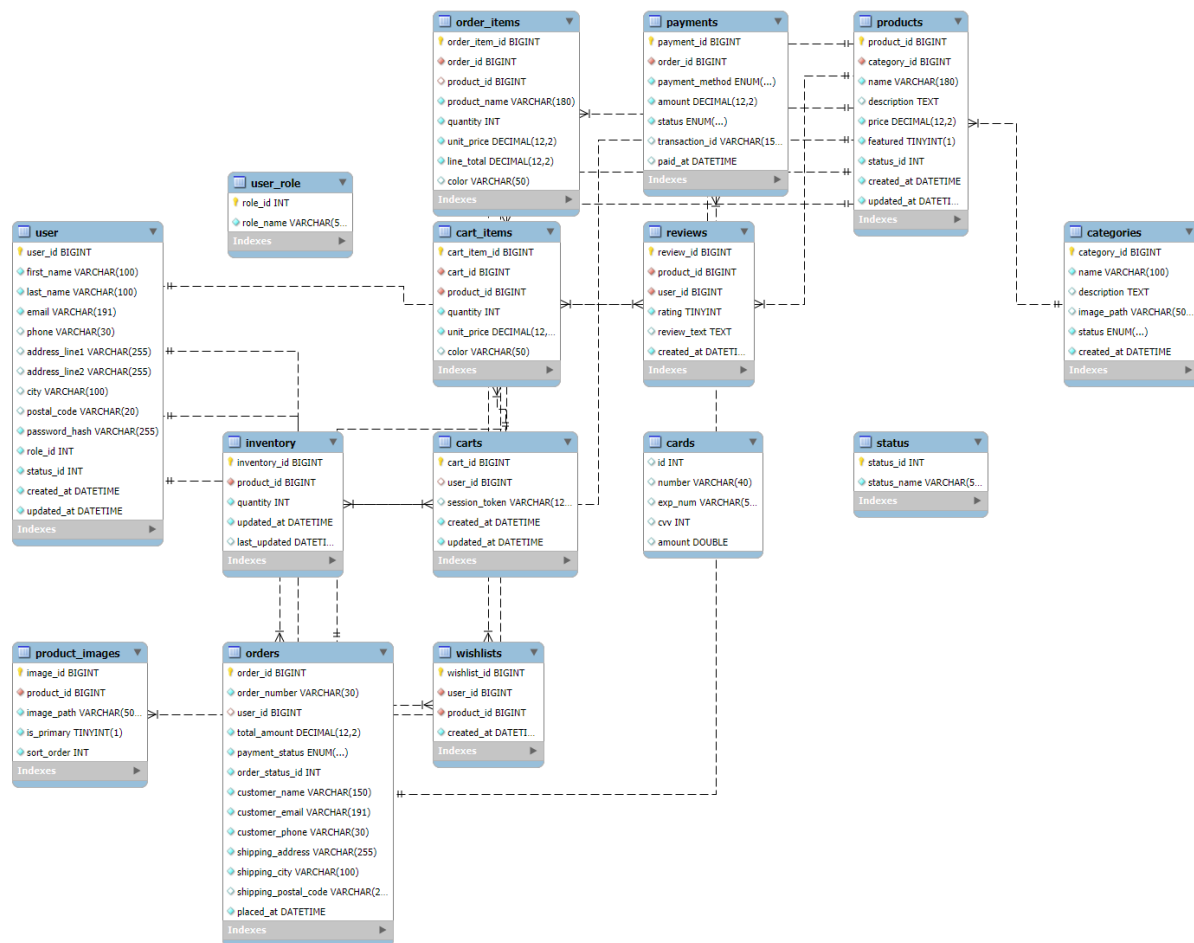


Figure 1: Database / ER Diagram

2.6 Folder Structure

The project follows a structured folder organization separating different types of files and functionality.

A typical project structure includes:

```

C:\.
  .css
  .js
  .php
  about.php
  addReviewProcess.php
  addToCartProcess.php
  adminLoginProcess.php
  cart.php
  cartProcess.php
  checkout.php
  checkoutProcess.php
  connection.php
  contact.php
  contactProcess.php
  faq.php
  footer.php
  header.php
  index.php
  loadProductsProcess.php
  login.php
  loginProcess.php
  logout.php
  product-detail.php
  README.md
  register.php
  registerProcess.php
  shop.php
  toggleWishlistProcess.php

  admin
    addProductProcess.php
    admin-dashboard.php
    admin-logout.php
    admin-orders.php
    admin-products.php
    admin-settings.php
    admin-users.php
    changeAdminPasswordProcess.php
    deleteProductProcess.php
    getProductProcess.php
    updateAdminAccountProcess.php
    updateOrderStatusProcess.php
    updateProductProcess.php
    updateUserRoleProcess.php

```

```

  assets
    css
      style.css
    js
      script.js

  Images
    background.png

    Category
      bed.png
      chair.png
      office.png
      sofa.png
      table.png

    collection
      Background-1.png
      Background-2.png
      Background-3.png
      Background-4.png
      Background.png

    featured
      Brass_desk.png
      Linear_Oak.png
      nordic_lounge.png
      sculptural_coffee_table.png

    mainproduct
      whitechair (1).png
      whitechair (2).png
      whitechair (3).png
      whitechair (4).png
      whitechair (5).png

    pairswell
      Background-1.png
      Background-2.png
      Background-3.png
      Background.png

```

```

  products
    aura_floor_lamp.png
    elysian_oak_table.png
    linear_sideboard.png
    nordic_lounge.png
    obelisk_sofa.png
    prod_6ac007ab3e39c_0.jpg
    prod_6ac007ab3fb5b_1.jpg
    serene_sleep.png

  wishlist
    Background-1.png
    Background.png
    chair.png

  PHPMailer
    Exception.php
    PHPMailer.php
    SMTP.php

  user
    dashboard.php
    invoice.php
    orders.php
    passwordChange.php
    removeWishlistProcess.php
    settingProcess.php
    settings.php
    wishlist.php

```

Figure 2: Project Folder Structure

3. SYSTEM IMPLEMENTATION

3.1 Home Page

The home page is the first interface presented to users.

It contains:

- Navigation menu.
- Brand identity.
- Hero section.
- Main call-to-action buttons.
- Featured products.
- Furniture categories.
- Customer review section.

The hero section introduces the main purpose of the website and provides buttons such as “Shop Now” and “Explore Categories”.

The design uses a modern furniture-oriented visual style with a focus on product presentation and ease of navigation.

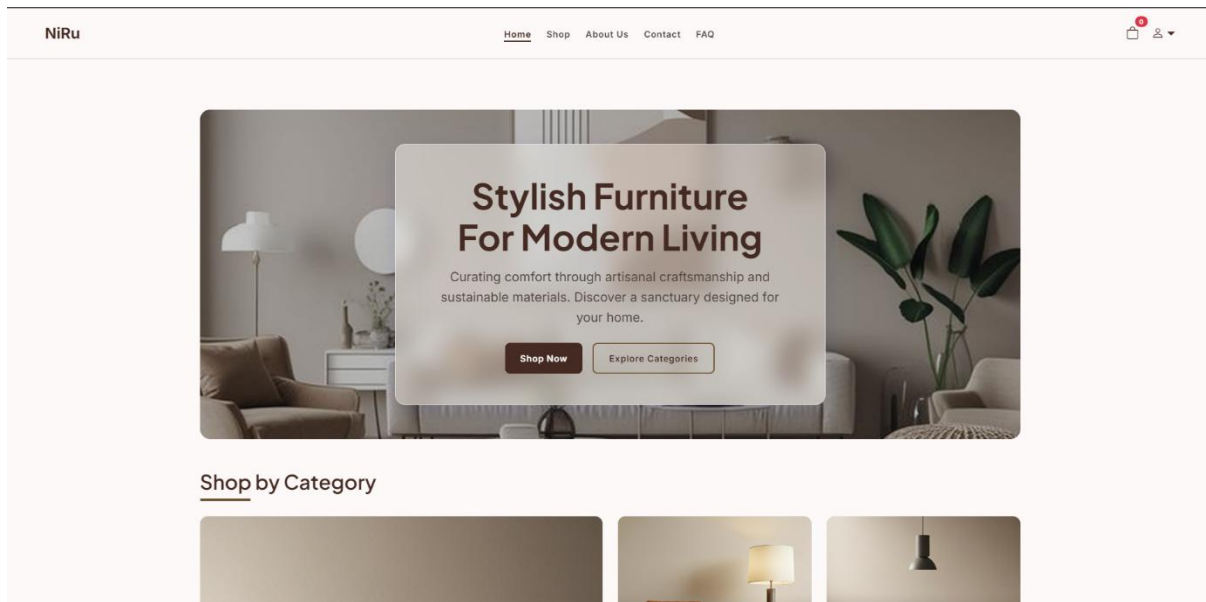


Figure 3: NiRu Furnitures Home Page

3.2 Shop Page

The Shop page provides access to the furniture catalogue.

The page includes:

- Product search.
- Category filters.
- Price-related filtering.
- Sorting options.
- Product cards.
- Product images.
- Product prices.
- Product ratings.
- View Details buttons.

The filtering and search functionality help users find relevant products more efficiently.

The submitted project document describes a left-side filtering area, keyword search, sorting options, and a main product grid.

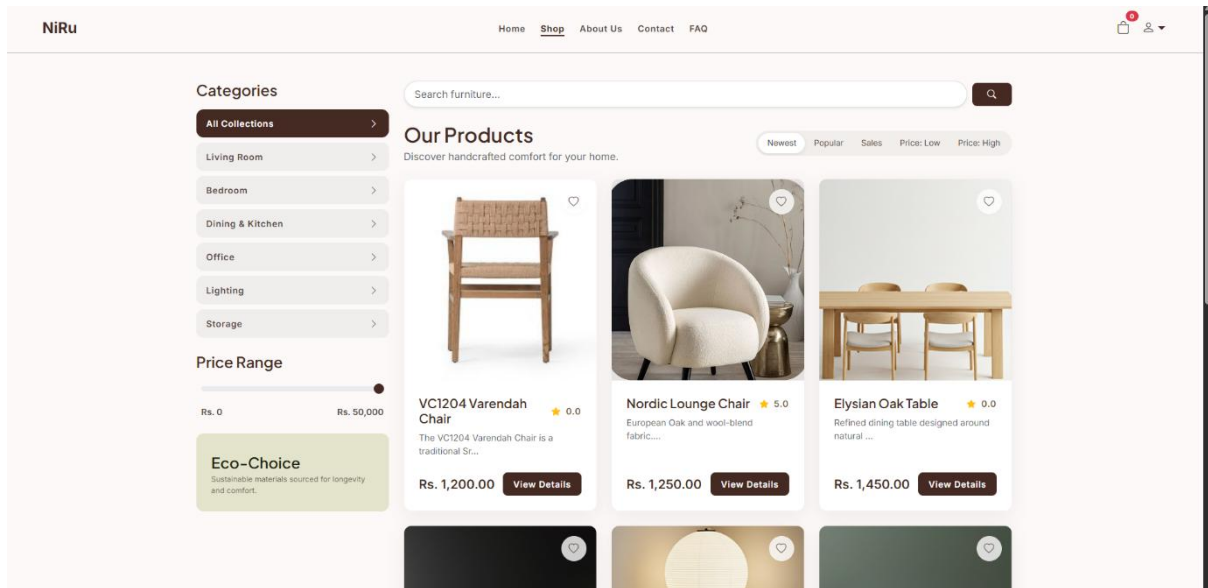


Figure 4: Shop Page

3.3 Product Details Page

The Product Details page provides complete information about an individual product.

The interface contains:

- Product image gallery.
- Product thumbnails.
- Product name.
- Product price.
- Product description.
- Color options.
- Quantity selector.
- Add to Cart button.
- Buy Now button.
- Shipping information.
- Customer reviews.

This page allows users to examine a product before deciding whether to purchase it.

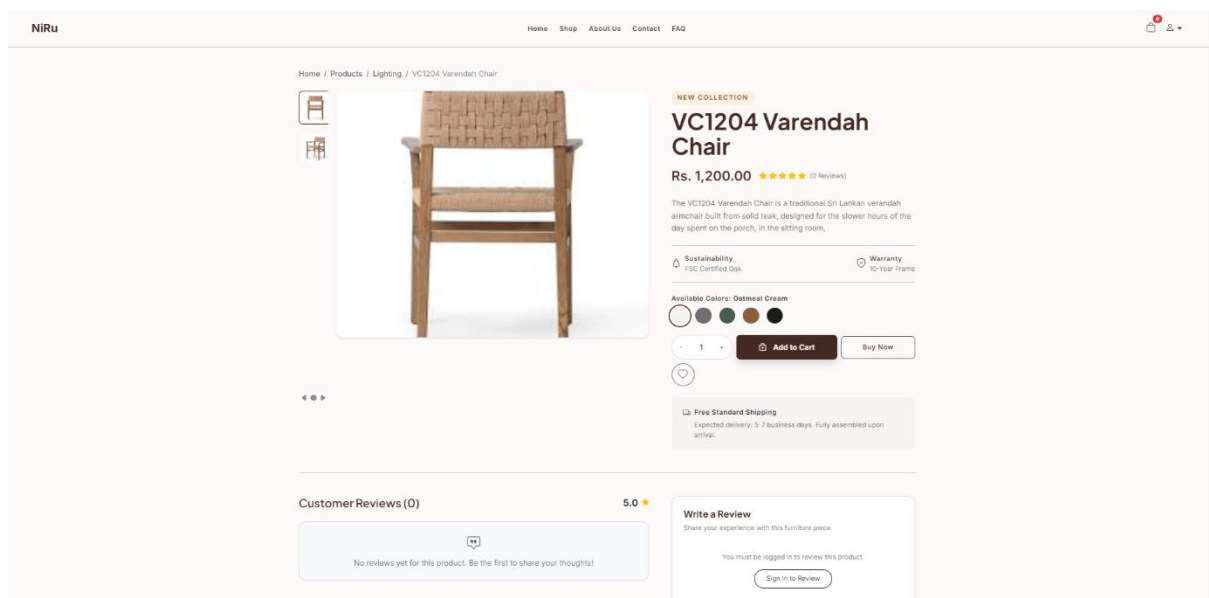


Figure 5: Product Details Page

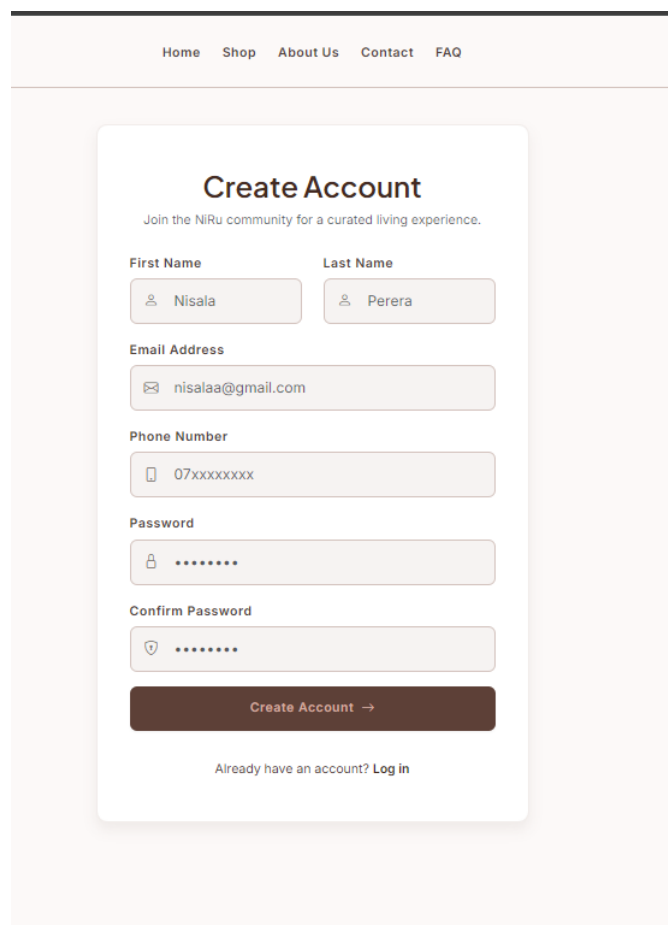
3.4 User Registration

The registration page allows new users to create accounts.

The registration form collects information such as:

- First name.
- Last name.
- Email address.
- Phone number.
- Password.
- Password confirmation.

The system validates the submitted information before creating the account.



The image shows a 'Create Account' form on a website. At the top, there is a navigation bar with links: Home, Shop, About Us, Contact, and FAQ. The form itself is titled 'Create Account' and includes a subtext: 'Join the NiRu community for a curated living experience.' The form fields are: First Name (with a person icon and the text 'Nisala'), Last Name (with a person icon and the text 'Perera'), Email Address (with an envelope icon and the text 'nisalaa@gmail.com'), Phone Number (with a phone icon and the text '07xxxxxxx'), Password (with a lock icon and seven dots), and Confirm Password (with a shield icon and seven dots). Below the fields is a dark brown button with the text 'Create Account →'. At the bottom of the form, there is a link: 'Already have an account? Log in'.

Figure 6: Create Account Page

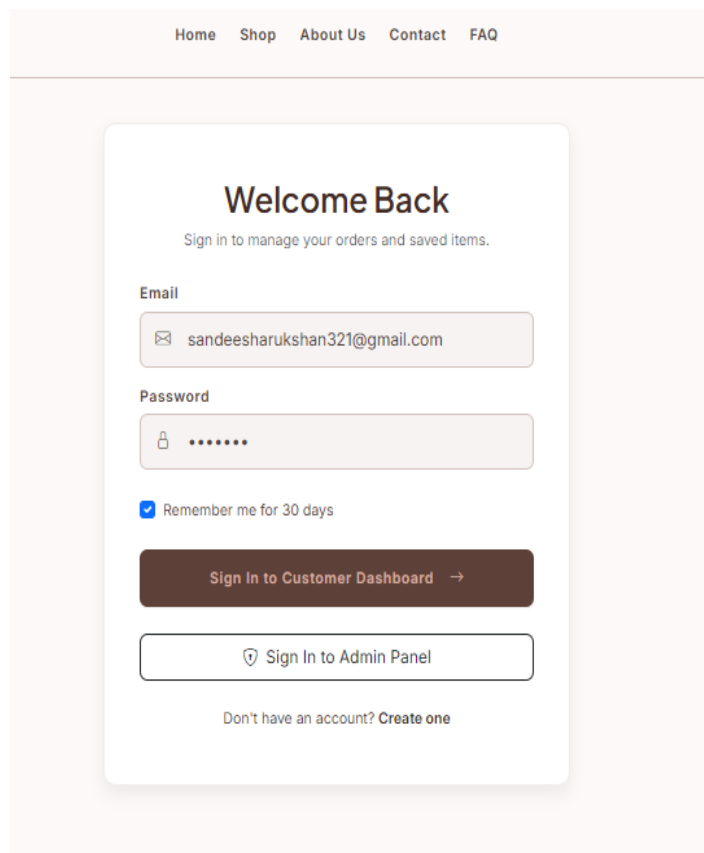
3.5 User Login

Registered users can access the system through the login page.

The login interface provides:

- Email input.
- Password input.
- Remember-me option.
- Login functionality.
- Registration link.

After successful authentication, users can access the appropriate customer or administrator area.



The image shows a login page with a light beige background. At the top, there is a navigation bar with links: Home, Shop, About Us, Contact, and FAQ. The main content area is a white card with a shadow. The card has the heading "Welcome Back" and a subtext "Sign in to manage your orders and saved items." Below this, there are two input fields: "Email" with a placeholder "sandeesharukshan321@gmail.com" and "Password" with a placeholder "*****". There is a checkbox labeled "Remember me for 30 days" which is checked. Below the inputs, there are two buttons: a dark brown button labeled "Sign In to Customer Dashboard →" and a white button with a dark border labeled "Sign In to Admin Panel". At the bottom of the card, there is a link "Don't have an account? Create one".

Figure 7: Login Page

3.6 Customer Dashboard

The Customer Dashboard provides customers with a centralized view of their account.

The dashboard includes:

- User profile information.
- Total orders.
- Orders in transit.
- Total spending.
- Wishlist count.
- Recent orders.
- Wishlist information.
- Navigation to account-related pages.

The submitted project report describes these dashboard components in detail.

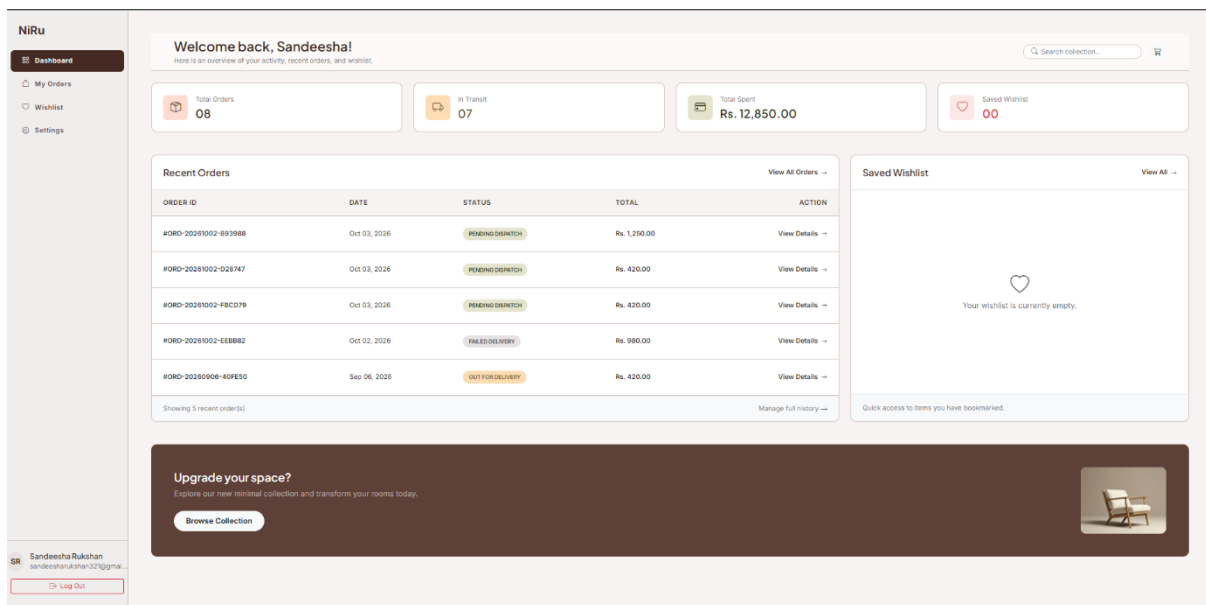


Figure 8: Customer Dashboard

3.7 Order Management

The My Orders page allows customers to view their order history.

- The page contains:
- Order IDs.
- Purchase dates.
- Order status.
- Order totals.
- Search functionality.
- Filtering options.
- View Details functionality

Example statuses include:

- Pending Dispatch
- Packed
- Out for Delivery

This provides users with a convenient method of monitoring their previous and current orders.

NIRu

Dashboard

My Orders

Wishlist

Settings

My Orders

Search orders... Filter

Total Orders: 08

In Transit: 07

Total Spent: Rs. 12,850.00

ORDER ID	DATE	STATUS	TOTAL AMOUNT	ACTIONS
ORD-20261002-893988	Oct 03, 2026	PENDING DISPATCH	Rs. 1,350.00	View Details →
ORD-20261002-028747	Oct 03, 2026	PENDING DISPATCH	Rs. 420.00	View Details →
ORD-20261002-FBCD79	Oct 03, 2026	PENDING DISPATCH	Rs. 420.00	View Details →
ORD-20261002-EBB882	Oct 02, 2026	FAILED DELIVERY	Rs. 980.00	View Details →
ORD-20260906-40FE50	Sep 06, 2026	OUT FOR DELIVERY	Rs. 420.00	View Details →
ORD-20260906-23489F	Sep 06, 2026	PENDING DISPATCH	Rs. 1,850.00	View Details →
ORD-20260906-03C318	Sep 06, 2026	PACKED	Rs. 5,660.00	View Details →
ORD-20260906-E4FE38	Sep 06, 2026	PENDING DISPATCH	Rs. 1,850.00	View Details →

Showing 8 of 8 orders

Upgrade your space?
Explore our new minimal collection and transform your rooms today.

Browse Collection

Sandeepa Rukshan
sandeeparukshan32@gmail.com
Log Out

Figure 9: Customer Order Management

3.8 Wishlist

The Wishlist allows customers to save products for future consideration.

The Wishlist provides:

- Saved product cards.
- Product images.
- Product prices.
- Add to Cart functionality.
- Delete functionality.
- Discover More option.

This feature allows users to maintain a collection of products they are interested in.

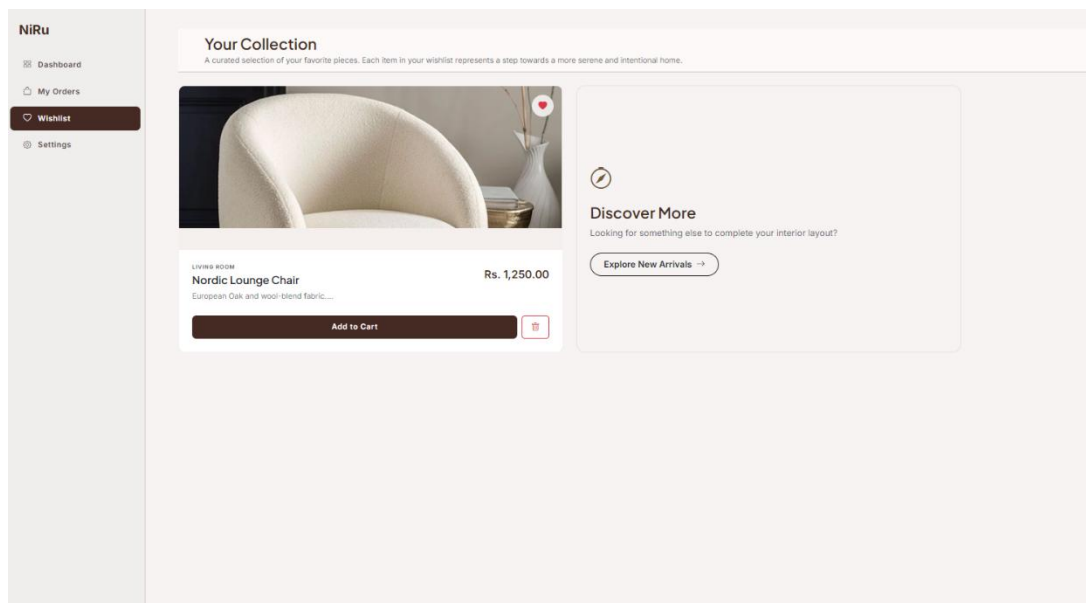


Figure 10: Wishlist

3.9 Account Settings

The Settings page allows customers to manage their personal information.

The page provides:

- Account information.
- Contact details.
- Shipping address.
- City.
- Postal code.

- Change password functionality.

The password-management functionality allows users to update their account credentials.

The screenshot displays the 'Settings' page of the NiRu application. On the left is a sidebar with navigation links: Dashboard, My Orders, Wishlist, and Settings (which is highlighted). The main content area is titled 'Settings' and contains two primary sections. The first section, 'Account Information', includes input fields for First Name (Sandeesh), Last Name (Rukshan), Phone Number (0713218157), Address Line 01 (E2-612/B, NEW TOWN), Address Line 02 (Optional) (EMBILIPITIYA), City (Embilipitiya), and Postal Code (70200). An 'Update Account' button is positioned to the right of these fields. The second section, 'Change Password', features a lock icon and input fields for Current Password, New Password, and Confirm New Password, all masked with dots. A 'Save Password' button is located at the bottom right of this section. At the bottom of the sidebar, the user's profile is shown with the initials 'SR', the name 'Sandeesh Rukshan', the email 'sandeeshrukshan321@gmail...', and a 'Log Out' button.

Figure 11: Account Settings

3.10 Admin Dashboard

The Admin Dashboard provides administrators with an overview of the system.

The dashboard contains KPI cards displaying:

- Total Products.
- Total Orders.
- Total Customers.
- Total Revenue.

It also includes a recent order table containing information such as order ID, customer details, order date, amount, and fulfillment status.

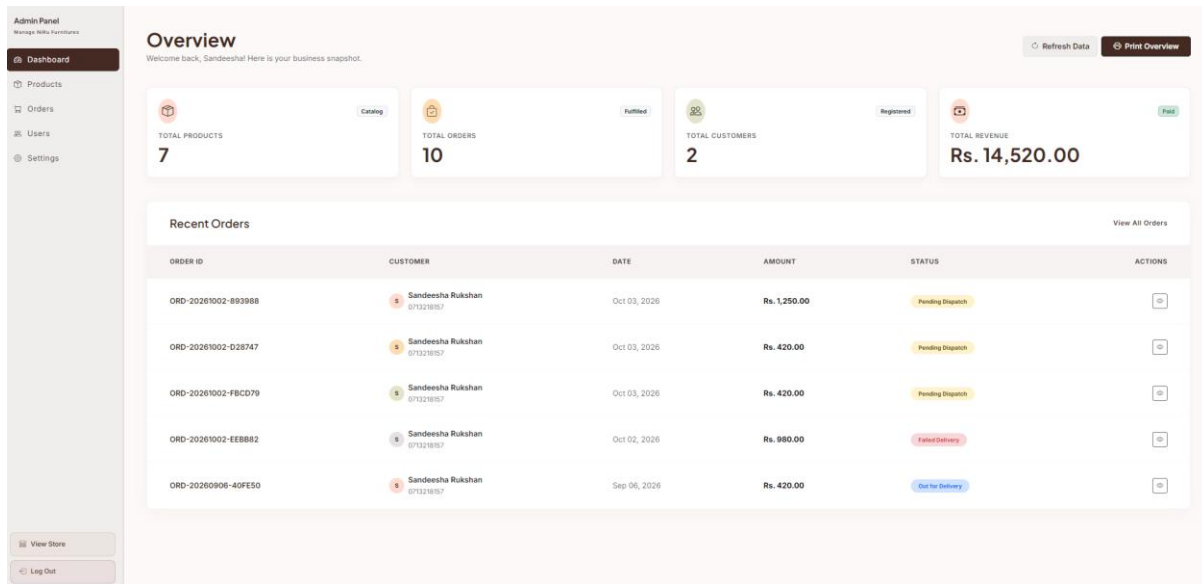


Figure 12: Admin Dashboard

3.11 Product Management

The administrator can manage furniture products through the Products page.

The system provides:

- Product listing.
- Product search/filtering.
- Product status filtering.
- Add Product functionality.
- Edit Product functionality.
- Delete Product functionality.
- Stock information.
- Product category information.
- Publication status.

The submitted project documentation describes filters for All, Active, and Draft/Inactive products as well as product management actions.

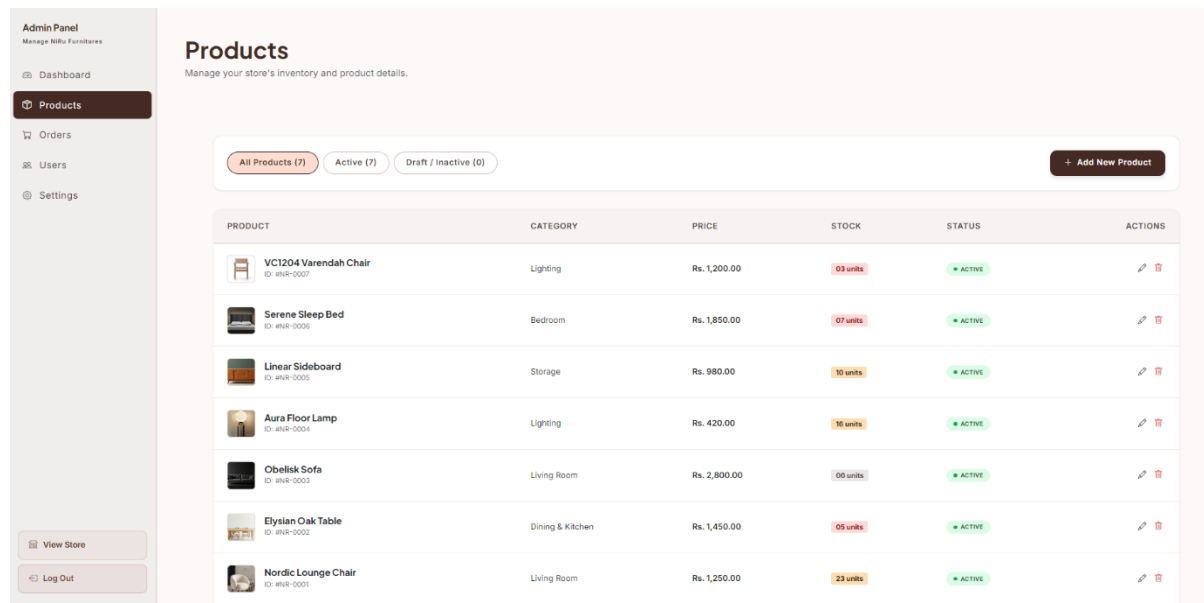


Figure 13: Product Management

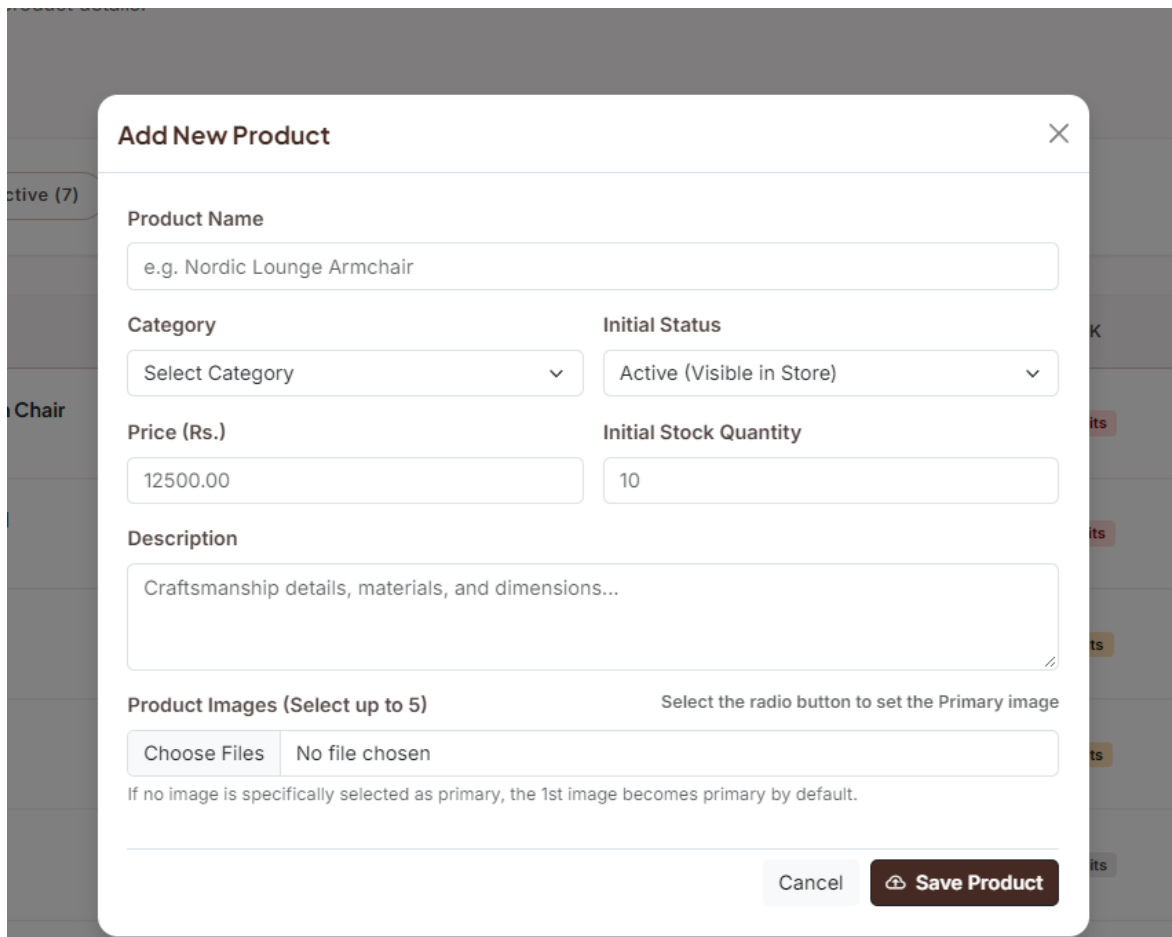
3.12 Add New Product

Administrators can create new product records using the Add New Product interface.

The form contains fields such as:

- Product name.
- Category.
- Visibility status.
- Price.
- Inventory quantity.
- Product description.
- Product images.

The administrator can upload multiple images and select the primary display image.

A screenshot of a web application's 'Add New Product' modal form. The form is white with a dark grey border and a close button (X) in the top right corner. It contains several input fields: 'Product Name' with a placeholder 'e.g. Nordic Lounge Armchair', 'Category' with a dropdown menu showing 'Select Category', 'Initial Status' with a dropdown menu showing 'Active (Visible in Store)', 'Price (Rs.)' with the value '12500.00', and 'Initial Stock Quantity' with the value '10'. There is a 'Description' text area with the placeholder 'Craftsmanship details, materials, and dimensions...'. Below this is a section for 'Product Images (Select up to 5)' with a note 'Select the radio button to set the Primary image'. It includes a 'Choose Files' button and a 'No file chosen' button. A small note below states 'If no image is specifically selected as primary, the 1st image becomes primary by default.' At the bottom right are 'Cancel' and 'Save Product' buttons.

Add New Product ✕

Product Name
e.g. Nordic Lounge Armchair

Category ▼ **Initial Status** ▼
Select Category Active (Visible in Store)

Price (Rs.) **Initial Stock Quantity**
12500.00 10

Description
Craftsmanship details, materials, and dimensions...

Product Images (Select up to 5) Select the radio button to set the Primary image
Choose Files No file chosen

If no image is specifically selected as primary, the 1st image becomes primary by default.

Cancel Save Product

Figure 14: Add New Product

3.13 Admin Order Management

The Admin Order Management page allows administrators to monitor and manage customer transactions.

The page includes:

- Total orders.
- Pending dispatch.
- Orders in process.
- Month-to-date revenue.
- Search.
- Status filtering.

- Order table.
- Customer information.
- Product information.
- Quantity.
- Total amount.
- Order date.
- Status.
- View/Edit controls.

The report documentation identifies these functions as part of the administrative order-management system

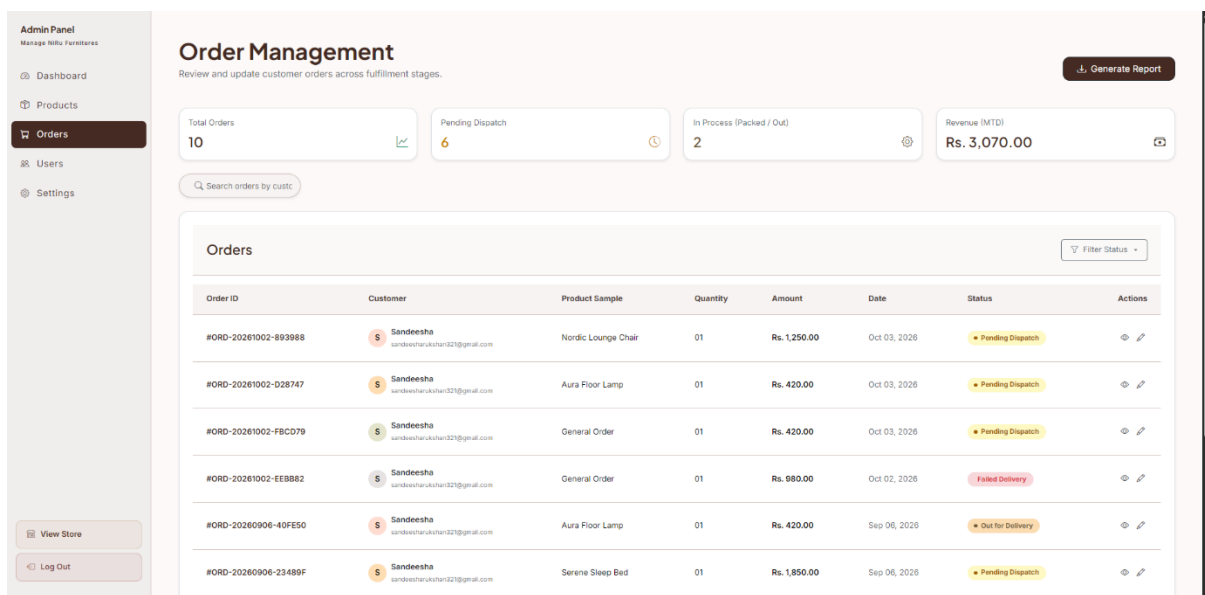


Figure 15: Admin Order Management

3.14 User and Role Management

The User and Role Management page allows administrators to manage registered users.

The page provides:

- User search.
- Role filtering.
- User ID.
- User name.
- Email address.
- Join date.
- Account status.

- Assigned role.
- Edit Role functionality.

The system distinguishes between Customer and Admin roles.

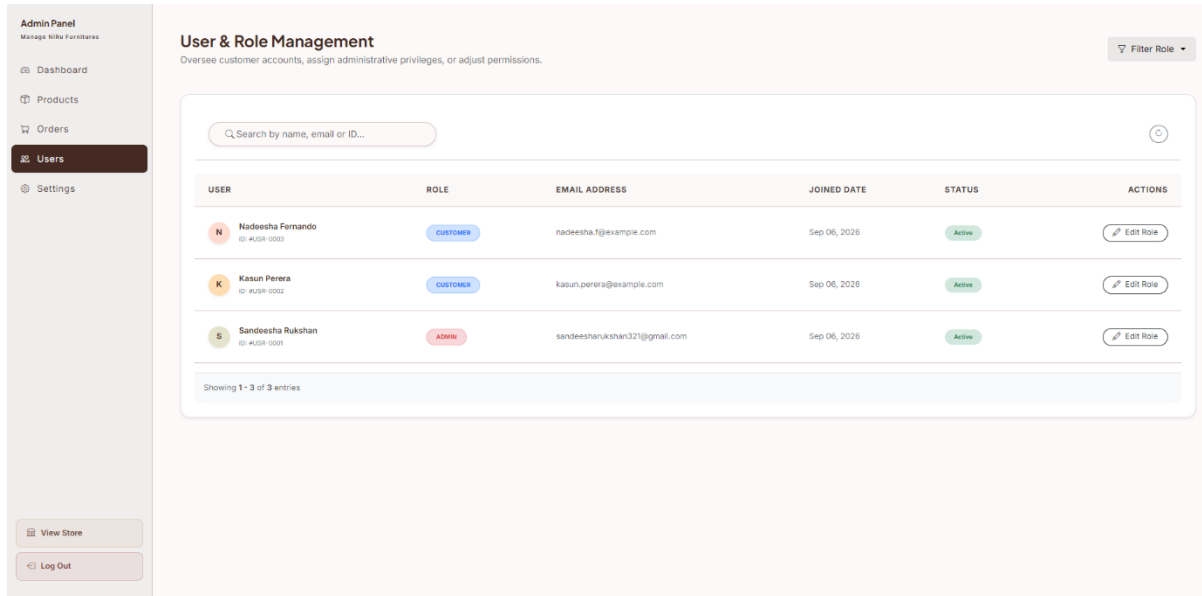


Figure 16: User and Role Management

3.16 Responsive Design

Responsive design was implemented to ensure that the application can adapt to different screen sizes.

The system is designed to work across:

- Desktop computers.
- Tablets.
- Mobile devices.

Bootstrap 5 was used to support responsive layouts through its grid system and utility classes.

The official project requirements specifically require the frontend to adapt to mobile, tablet, and desktop screen sizes.

4. CHALLENGES AND SOLUTIONS

Challenge

One of the major challenges was maintaining consistent layouts across different screen sizes. Navigation bars, sidebars, product grids, and other interface elements could overlap or become difficult to use on smaller screens.

Solution

Bootstrap 5 was integrated into the project to provide a responsive grid system and reusable utility classes.

This reduced the need for excessive custom media queries and helped maintain consistent layouts across different devices.

The submitted project documentation specifically identifies responsive layout problems and the use of Bootstrap 5 as the solution.

4.2 Static and Dynamic Content Integration

Challenge

Another challenge was integrating PHP-generated database information into previously designed HTML layouts.

Directly inserting PHP loops and dynamic data into static markup could make the HTML difficult to maintain.

Solution

Alternative PHP control syntax was used to maintain readable HTML structures.

Dynamic output was also sanitized using `htmlspecialchars()` to reduce the possibility of HTML and Cross-Site Scripting-related problems.

4.3 User Session Management

Challenge

Protecting private customer pages was an important challenge.

Unauthenticated users should not be able to access customer dashboards or other protected pages.

Solution

PHP session management was implemented using `session_start()`.

Protected routes verify the relevant session information before allowing access. Users who are not authenticated are redirected to the login page.

The submitted report describes session validation and route protection as part of the implemented solution.

4.4 Data Security and Input Handling

The project also required careful handling of user-generated data.

Input validation and data sanitization were applied to reduce security risks and prevent unexpected content from affecting the interface.

The project documentation identifies the use of `htmlspecialchars()` for sanitizing dynamic output.

The official ICT 1209 guide also requires prepared statements through PDO or MySQL to reduce SQL injection risks.

5. INDIVIDUAL CONTRIBUTION

5.1 Backend Developer

M.P. Sandeesha Rukshan
ITT/2024/092 – 2777

The backend developer was responsible for the server-side functionality and database-related components of the project.

Major contributions included:

Database Architecture and Schema Design

- Designed the MySQL relational database.
- Created and maintained database tables.
- Managed relationships between users, products, categories, orders, and wishlists.
- Ensured consistency of stored information.

PHP Backend Development

- Developed server-side PHP functionality.
- Connected the application to the database.
- Implemented dynamic data retrieval.
- Processed customer orders.
- Processed user interactions.

Authentication and Access Control

- Implemented login functionality.
- Implemented session handling.
- Protected private pages.
- Distinguished customer and administrator roles.
- Redirected unauthorized users.

CRUD and Inventory Management

Developed functionality for:

- Creating products.
- Reading product information.
- Updating products.
- Deleting products.
- Monitoring stock levels.
- Managing order statuses.

Security and Data Sanitization

- Implemented input validation.
- Applied output sanitization.
- Used `htmlspecialchars()` where appropriate.
- Worked on preventing security problems caused by unsafe dynamic data.

These contributions correspond to the backend responsibilities documented in the project report.

5.2 Frontend Developer

Nisala Anjana

ITT/2024/027 – 2712

The frontend developer was responsible for designing and implementing the user interface and visual presentation of the application.

UI/UX Layout and Prototyping

Designed and structured the interfaces for:

- Storefront.
- Product details.
- Customer dashboard.
- Admin management panels.
- Authentication pages.
- Forms and other user interfaces.

Responsive Styling

Bootstrap 5 was used to make the application responsive across:

- Desktop.
- Tablet.
- Mobile

Responsive components included:

- Navigation menus.
- Product cards.
- Tables.
- Dashboard layouts.
- Forms.

Visual Presentation

Responsible for:

- Branding.
- Colour schemes.
- Image galleries.
- Product presentation.
- Navigation states.
- Status badges.
- Overall visual consistency.

Form and Modal Design

Designed and implemented:

- Login forms.
- Registration forms.
- Profile forms.
- Product creation forms.
- Product-management modals.

The submitted project documentation identifies these responsibilities under the frontend developer contribution.

6. SYSTEM REQUIREMENTS AND SETUP INSTRUCTIONS

6.1 Software Requirements

The system requires:

- XAMPP.
- Apache.
- MySQL.
- PHP.
- A modern web browser.
- Code editor.

The project was designed to operate within a local XAMPP environment.

6.2 Deploy Project Files

First, place the project inside the XAMPP htdocs directory.

For example:

C:\xampp\htdocs\nirufurnitures

The official project documentation specifies that the project files should be placed inside the XAMPP root directory and accessed through the local server.

6.3 Start XAMPP

1. Open the XAMPP Control Panel.
2. Start Apache.
3. Start MySQL.
4. Confirm that both services are running.

The services should show an active/running status.

6.4 Configure Database Connection

Open the database connection file:

connection.php

Configure the database connection according to the local MySQL configuration.

Example:

- `$servername = "localhost";`
- `$username = "root";`
- `$password = "yourpassword";`
- `$dbname = "niru_furniture";`

The exact connection configuration should match the local MySQL environment.

6.5 Import the Database

Open a browser and navigate to:

<http://localhost/phpmyadmin/>

Then:

- Select **New**.
- Create the database:

niru_furniture

1. Select the newly created database.
2. Open the **Import** tab.
3. Select the project's SQL file.
4. Click **Import**.

The project documentation specifies these database-import steps.

6.6 Launch the Application

After starting Apache and MySQL and importing the database, open:

<http://localhost/nirufurnitures/>

The NiRu Furnitures application should then load and communicate with the local MySQL database.

7. CONCLUSION

NiRu Furnitures was developed as a database-driven furniture showcase and order management web application.

The system provides separate functionality for customers and administrators. Customers can browse furniture products, search and filter products, view detailed information, register and log into the system, manage their profiles, maintain wishlists, and monitor their orders.

Administrators are provided with a centralized management interface through which they can manage products, inventory, orders, customers, and user roles.

The project combines frontend technologies including HTML5, CSS3, Bootstrap 5, and JavaScript with PHP and MySQL to create an interactive web application. The official ICT 1209 project requirements specify these technologies as the required technology stack.

During development, several technical challenges were encountered, particularly in responsive design, dynamic PHP integration, session management, and data handling. These challenges were addressed through Bootstrap's responsive utilities, structured PHP development, session-based access control, and data sanitization.

The project also provided practical experience in web application development, database integration, user authentication, responsive interface design, CRUD operations, and collaborative software development.

Overall, the completed system demonstrates the practical application of the concepts covered in ICT 1209 – Web Technologies and provides a functional foundation for a furniture-oriented e-commerce and order management platform.

8. REFERENCES

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